

Linear Regression – Apartment Rent Prediction

CODING :

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data = {
    "Area": [600, 800, 1000, 1200, 1500, 1800, 2000, 2200],
    "Rooms": [1, 2, 2, 3, 3, 4, 4, 5],
    "Location": [1, 1, 2, 2, 3, 3, 4, 4],
    "Rent": [10000, 13000, 16000, 19000, 23000, 27000, 30000, 33000]
}

df = pd.DataFrame(data)

X = df[["Area", "Rooms", "Location"]]
y = df["Rent"]
model = LinearRegression()
model.fit(X, y)

new_apartment = [[1400, 3, 3]]
prediction = model.predict(new_apartment)

print("LINEAR REGRESSION")
print("-----")
print("Area Coefficient:", round(model.coef_[0], 2))
print("Rooms Coefficient:", round(model.coef_[1], 2))
print("Location Coefficient:", round(model.coef_[2], 2))
print("Intercept:", round(model.intercept_, 2))

print("\nPredicted Rent:", round(prediction[0], 2))
```

OUTPUT :

```
LINEAR REGRESSION
-----
Area Coefficient: 10.0
Rooms Coefficient: 1000.0
Location Coefficient: 1000.0
Intercept: 2000.0

Predicted Rent: 22000.0
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: Us
warnings.warn(
```

2. Linear Classification – Spam Email Detection

CODING :

```
import numpy as np
from sklearn.linear_model import LogisticRegression
X = np.array([
    [5, 4],
    [4, 3],
    [6, 5],
    [1, 0],
    [0, 1],
    [2, 1],
    [5, 3],
    [1, 1]
])
y = np.array([1, 1, 1, 0, 0, 0, 1, 0])

model = LogisticRegression()
model.fit(X, y)
new_email = [[3, 2]]
prediction = model.predict(new_email)
print("LINEAR CLASSIFICATION")
print("-----")
print("Coefficients:", model.coef_)
print("Intercept:", model.intercept_)

if prediction[0] == 1:
    print("New Email: SPAM")
else:
    print("New Email: NON-SPAM")
```

OUTPUT:

```
LINEAR CLASSIFICATION
-----
Coefficients: [[0.90713108 0.69710396]]
Intercept: [-4.16027626]
New Email: NON-SPAM
```

3. Naïve Bayes – Flu Prediction

CODING :

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
data = {
    "Fever":    [1, 1, 0, 1, 0, 1, 0, 1],
    "Cough":    [1, 0, 1, 1, 0, 1, 0, 0],
    "Headache": [1, 1, 1, 0, 0, 1, 0, 1],
    "Flu":      [1, 1, 0, 1, 0, 1, 0, 1]
}
df = pd.DataFrame(data)
X = df[["Fever", "Cough", "Headache"]]
y = df["Flu"]
model = GaussianNB()
model.fit(X, y)
new_patient = [[1, 1, 0]]
prediction = model.predict(new_patient)
probability = model.predict_proba(new_patient)
print("NAIVE BAYES CLASSIFICATION")
print("-----")
print("Prior Probabilities:")
print("No Flu :", round(model.class_prior_[0], 2))
print("Flu    :", round(model.class_prior_[1], 2))
print("\nNew Patient:")
print("Fever = 1, Cough = 1, Headache = 0")
if prediction[0] == 1:
    print("Prediction: FLU")
else:
    print("Prediction: NO FLU")
print("\nProbability of No Flu:", round(probability[0][0], 4))
print("Probability of Flu:", round(probability[0][1], 4))
```

OUTPUT:

```
NAIVE BAYES CLASSIFICATION
-----
Prior Probabilities:
No Flu : 0.38
Flu    : 0.62

New Patient:
Fever = 1, Cough = 1, Headache = 0
Prediction: FLU

Probability of No Flu: 0.0
Probability of Flu: 1.0
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning
```

